

# Infrastructure & Commands

Paths, Docker, Config, Post-Start, Persistence

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ClawCraft Project Reference

# Architecture Overview

```
Discord (#blender, @ClawBot)
-> OpenClaw Gateway (Docker, port 18789)
-> GPT-5.3-Codex (free via ChatGPT sub)
-> Blender Plugin (TypeScript, TCP socket)
-> Blender 5.1 addon (host.docker.internal:9876)
-> EEVEE render -> PNG -> shared volume
-> MEDIA: directive -> Discord
```

## Key Paths -- Mac (Host)

- **Blender 5.1** /Applications/Blender 3.app
- **Blender 4.5.2 fallback** /Applications/Blender.app
- **Blender MCP addon** ~/Library/Application Support/Blender/5.1/scripts/addons/blender\_mcp\_addon.py
- **Plugin source** ~/.openclaw/extensions/blender/
- **index.ts** Entry point, registers blender/canvas/slides tools
- **src/blender-tool.ts** Core tool: scene\_info, execute, render, Notion push
- **src/blender-socket.ts** TCP socket client for Blender addon
- **src/canvas-tool.ts** HTML -> Chromium -> PNG
- **src/slides-tool.ts** Markdown -> Slides -> screenshots
- **package.json** @openclaw/blender, type: module, no deps
- **OpenClaw source** ~/src/openclaw/
- **Docker compose** ~/src/openclaw/docker-compose.yml
- **Docker-compose .env** ~/src/openclaw/.env (THIS is where env vars go)
- **OpenClaw config** ~/.openclaw/openclaw.json
- **Workspace** ~/.openclaw/workspace/
- **Post-start script** ~/src/openclaw/docker-post-start.sh
- **Render output** /tmp/clawcraft/
- **ClawDock helpers** ~/.clawdock/clawdock-helpers.sh
- **Logs archive** ~/clawcraft\_logs/
- **Fridge physics (separate)** /Users/princessa/fridge\_physics\_phase1.blend

## Key Paths -- Docker Container

Container: openclaw-openclaw-gateway-1. Image: openclaw:local.

- **Home** /home/node
- **Config** /home/node/.openclaw/openclaw.json
- **Workspace** /home/node/.openclaw/workspace/
- **Plugin** /home/node/.openclaw/extensions/blender/
- **Renders** /home/node/.openclaw/media/renders/
- **Sessions** /home/node/.openclaw/sessions.json

- **Auth profiles** /home/node/.openclaw/agents/main/agent/auth-profiles.json
  - **Gateway port** 18789
  - **Bridge port** 18790
  - **Extra port** 1455
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## Docker Compose Configuration

### **docker-compose.yml key settings:**

- Image: \${OPENCLAW\_IMAGE:-openclaw:local}
- env\_file: .env (reads from ~/src/openclaw/.env)
- Volume: \${OPENCLAW\_CONFIG\_DIR}:/home/node/.openclaw (config + plugins)
- Volume: \${OPENCLAW\_WORKSPACE\_DIR}:/home/node/.openclaw/workspace
- Volume: /tmp/clawcraft:/home/node/.openclaw/media/renderers (shared renders)
- extra\_hosts: host.docker.internal:host-gateway
- Ports: 18789 (gateway), 18790 (bridge), 1455
- restart: unless-stopped

### **Environment variables in .env:**

- OPENCLAW\_CONFIG\_DIR=/Users/princessa/.openclaw
  - OPENCLAW\_WORKSPACE\_DIR=/Users/princessa/.openclaw/workspace
  - OPENCLAW\_GATEWAY\_PORT=18789
  - OPENCLAW\_GATEWAY\_BIND=lan
  - OPENCLAW\_GATEWAY\_TOKEN=a571f94f...
  - OPENCLAW\_IMAGE=openclaw:local
  - OPENAI\_API\_KEY=sk-proj-...
  - NOTION\_TOKEN=ntn\_...
  - NOTION\_RENDERERS\_DB=32f36619-2ab5-8149-af30-c540c854aff4
  - NOTION\_SCENES\_DB=32f36619-2ab5-8123-aedc-edeb780ff
  - NOTION\_TASKS\_DB=32f36619-2ab5-8166-82ec-f09730324655
  - NOTION\_MILESTONES\_DB=32f36619-2ab5-81cf-ab76-fc564d44fcf5
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## Persistence Model

### **Survives container recreate (Mac-mounted):**

- Plugin files (~/.openclaw/extensions/blender/)
- Workspace files (~/.openclaw/workspace/)
- Config (~/.openclaw/openclaw.json)
- Docker image (openclaw:local), docker-compose.yml, .env

- Render files (/tmp/clawcraft/)
- docker-post-start.sh

### Wiped on recreate (reinstalled by docker-post-start.sh, ~60s):

- apt: gh, ffmpeg, python3-pip, asciinema, build-essential
- pip: matplotlib, pillow, yt-dlp, nano-pdf, plotly
- npm: @slidev/cli, @slidev/theme-default, slidev-theme-light-icons, slidev-addon-python-runner, plotly.js-dist-min, @notionhq/client
- render-html.mjs (Playwright HTML->PNG script at /app/)
- Plugin file ownership fix (chown -R node:node)
- jiti cache clear (/tmp/jiti/src-blender-\*.cjs)
- Log file creation (render\_log.json, hand\_built\_log.json, workflow\_notes.md)
- Previous session log snapshot to ~/clawcraft\_logs/

## Common Commands

### Startup (after reboot)

```
# 1. Open Blender 5.1, sidebar N -> BlenderMCP -> Start Server
# 2. Start Docker container
cd ~/src/openclaw && docker compose up -d
# 3. Run post-start script
bash docker-post-start.sh
```

### Check status

```
# Container running?
docker ps --filter name=openclaw
# Blender socket reachable from Docker?
docker exec openclaw-openclaw-gateway-1 node -e \
  "const n=require('net');const s=new n.Socket();\
  s.setTimeout(3000);s.connect(9876,'host.docker.internal',\
  ()=>{console.log('connected');s.destroy()});\
  s.on('error',e=>console.log('FAILED:',e.message))"
# Blender listening on Mac?
lsof -nP -i :9876
# Container env vars?
docker exec openclaw-openclaw-gateway-1 env | grep NOTION
# Models status?
docker exec openclaw-openclaw-gateway-1 node /app/dist/index.js models list
docker exec openclaw-openclaw-gateway-1 node /app/dist/index.js models status
```

### Trigger agent (render via Discord)

```
docker exec openclaw-openclaw-gateway-1 node /app/dist/index.js \
  agent --to "discord:channel:1485731738343702628" \
  --message "Your prompt here" --deliver
```

### Send message AS bot (no AI trigger)

```
docker exec openclaw-openclaw-gateway-1 node /app/dist/index.js \  
  message send --channel discord \  
  --target "1485731738343702628" \  
  --message "Your message here"
```

## Switch models

```
# To Sonnet  
docker exec openclaw-openclaw-gateway-1 node /app/dist/index.js \  
  config set agents.defaults.model.primary \  
  anthropic/claude-sonnet-4-20250514  
docker exec openclaw-openclaw-gateway-1 sh -c \  
  'echo "{}" > /home/node/.openclaw/sessions.json'  
  
# To Codex  
docker exec openclaw-openclaw-gateway-1 node /app/dist/index.js \  
  config set agents.defaults.model.primary \  
  openai-codex/gpt-5.3-codex  
docker exec openclaw-openclaw-gateway-1 sh -c \  
  'echo "{}" > /home/node/.openclaw/sessions.json'
```

## View container logs

```
docker logs openclaw-openclaw-gateway-1 --tail 100  
# Or use helper:  
clawdock-logs
```

## Shell into container

```
docker exec -it openclaw-openclaw-gateway-1 bash  
# Or: clawdock-shell
```

## Nuke jiti cache (after plugin code changes)

```
docker exec openclaw-openclaw-gateway-1 rm -f /tmp/jiti/src-blender-*.cjs /tmp/jiti/src-canvas-*.cjs /tmp/jiti/
```